

WASP-135b

R-1192

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_220319 · created 2026-07-31T05:48:44+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459657.9357 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.013
Transit depth (Rp/Rs)^2	3.08 +/- 0.45 [%]
Orbital Inclination (inc)	81.9 +/- 1.48
Airmass coefficient 1 (a1)	1.027 +/- 0.024
Airmass coefficient 2 (a2)	-0.022 +/- 0.02
Scatter in the residuals of the lightcurve fit is	2.31 %
Best Comparison Star	#3 - [417, 258]
Optimal Method	PSF photometry
Transit Duration (day)	0.07 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.176 ± 0.013 vs 0.139 (deviation 2.81σ)
Duration fit vs published	0.07 d vs 0.0687 d (deviation 0.11σ)
Mid-time O–C	-14.0 min
Depth SNR	13.4
Residual scatter	2.31 %

Light curve

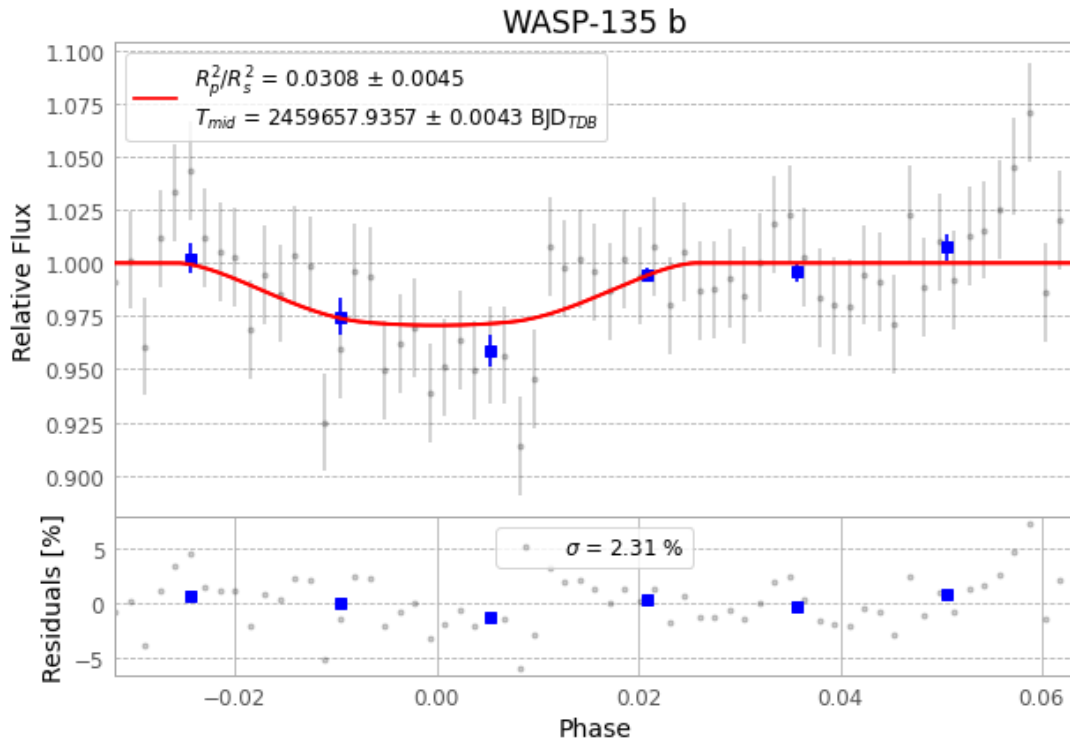


Figure 1 — FinalLightCurve_WASP-135 b_2022-03-19.png (SHA-256 fb6e67823b55bfbe...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [197, 257], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 57521472fd26c4d8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

65 FITS frames (43 MB), WASP-135220319092113.FITS → WASP-135220319123313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-220319.log (e50401ecf68c...), FinalLightCurve_WASP-135 b_2022-03-19.png (fb6e67823b55...), FinalLightCurve_WASP-135 b_2022-03-19.csv (398a40f916ab...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 05:48Z.